



Skull means: Skeleton of the head.

Skull functions are:

1. Enclose the brain.
2. Houses the organs of the special sense (seeing, hearing, tasting & smelling).
3. Surrounds the openings into digestive & respiratory tracts.

Anatomically the skull should be oriented so that inferior margin of orbit and superior margin of external acoustic meatus are horizontal, this line called Orbitomeatal plane (Frankfurt Plane).

Skull without mandible called Cranium.

Cranium is divided into 2 main parts:

A) Calvaria (brain box) (upper part):

1- Skull Cap. 2- Cranial Base.

B) Facial skeleton (lower part).

Method of study of skull:

A. Study of skull by using individual bones.

B. Study of skull as a whole:

1- External views. 2- Internal views.

External views are:

1. Norma Verticalis (superior view).
2. Norma Occipitalis (posterior view).
3. Norma Frontalis (anterior view).
4. Norma Lateralis.
5. Norma Basalis (inferior view).

NORMA VERTICALIS:



THE SKULL



L/ 1.

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It means viewing of skull from above.

Shape: oval wide posteriorly, narrow anteriorly.

Bones: Frontal, occipital & Parietal bones.

Sutures:

1. Coronal Suture: which lies between frontal bone anteriorly & two parietal bones posteriorly.
2. Sagittal Suture: which lies between two parietal bones.
3. Lambdoid Suture: which lies between occipital bone & two parietal bones.

The meeting point between coronal & sagittal sutures is called (Bregma), this point remains unossified from birth till (1.5year) (18 months). It is called Anterior Fontanel.

The meeting point between sagittal & lambdoid sutures is called (Lambda), this fuses at 3 months, it is called Posterior Fontanel.

Clinical Importance of Fontanels:

1. Depressed fontanel indicate dehydration.
2. Elevated fontanel indicate increase I.C.P.
3. Early closed indicate Microcephaly.
4. Delayed closed indicate Ricketts.

Vertex: is the highest point on sagittal suture.

Vault: is the arched roof or dome of skull.

Parietal foramen: it lies 1.5 inch in front of lambda.

Obelion: point on sagittal suture between two parietal foramina.

NORMA OCCIPITALIS:

It resembles abroad arch, flattened below, convex up.

Bones: Parietal, Occipital & Mastoid part of temporal bones.

Sutures:

1. Lambdoid suture: which lies bet. Occipital & Parietal bones.
2. Parietomastoid suture: which lies between Parietal bone & mastoid.
3. Occipitomastoid suture: which lies bet. Mastoid & occipital bone.



L/ 1.

THE SKULL

Dr. Munqith Mazin



Features:

- a) **External occipital protuberance** which is a prominent ridge represent the junction of head & neck. The most prominent point of this protuberance is called Inion. Trapezius attached to its upper part, while its lower part gives attachment to ligamentum nuchae.
- b) **Highest Nuchal Line:** it starts from upper part external occipital protuberance, it gives attachment to epicranial aponeurosis medially, and to occipitalis muscle laterally.
- c) **Superior Nuchal Line:** it started from ext. occipital protuberance, it extends laterally, it lies below the highest nuchal line (1 cm.) medially. It gives origin of trapezius laterally; it gives insertion of sternomastoid & splenius capitis.

NORMA FRONTALIS:

It is oval in shape, wide above than below.

Bones: Frontal bone, two Maxillae, two Nasal bones, Zygomatic bones & Mandible.

Features:

- a) **Superciliary Arch:** which is rounded, curved elevation lies just above medial part of orbit, it covers frontal sinus, it is more marked in males than females.
- b) **Glabella:** which is a median elevation connecting the two superciliary arches.
- c) **Nasion:** is a median point at the root of the nose, it is the site of meeting of frontonasal & internasal sutures.
- d) **Frontal Eminence:** which is a rounded elevation above the superciliary arch.
- e) **Orbit:** The eye ball surrounded by four-sided pyramidal bony orbit. The pyramid has base, apex & walls. The Base is the orbital opening which has four margins:
 - 1. Supra orbital margin: which is formed by frontal bone. At the junction of lateral $\frac{2}{3}$ & medial $\frac{1}{3}$ there is supra orbital notch (foramen).
 - 2. Infra orbital margin: formed by zygomatic bone laterally & maxilla medially.
 - 3. Lateral orbital margin: formed by frontal process of zygomatic bone & zygomatic process of frontal bone.
 - 4. Medial orbital margin: formed by frontal bone above & lacrimal crest of frontal process of maxilla below.
- f) **Anterior Nasal Aperture:** it is a pear – shaped, wide below and narrow above, bounded superiorly by lower border of nasal bone, inferiorly by body of maxilla.



THE SKULL



L/ 1.

Dr. Munqith Mazin

***Nasal bone** articulates with the opposite bone at (Inter nasal suture), and posteriorly with frontal process of maxilla, superiorly with frontal bone at fronto-nasal suture and inferiorly with nasal cartilage.

***Rhinion** is the lower part of inter nasal suture.

***Anterior Nasal Spine** is a sharp projection projects from lower border of nasal aperture.

g) **Maxilla:** It is a large bone, form large portion of face, it has the following:

1. Nasal notch medially
2. Anterior nasal spine
3. Infra orbital foramen below infra orbital margin
4. Incisive fossa
5. Canine fossa.

Maxilla has 3 processes:

1. Frontal process of maxilla; which articulates with nasal bone anteriorly & with lacrimal bone posteriorly & with frontal bone superiorly
2. Zygomatic process; articulate with zygomatic bone
3. Alveolar process; which bear sockets of upper teeth.

h) **Zygomatic Bone**

i) **Mandible**

NORMA LATERALIS:

Features:

- 1) Temporal Lines; it starts as single line at zygomatic process of frontal bone. It ascends upward & backward reach to parietal bone where it divided into superior & inferior line, the superior one fade out, but the inferior one continues downward & forward to become continuous with supra mastoid crest.
- 2) Zygomatic Arch; it is horizontal bar in front of ear, it is formed by temporal process of zygomatic and zygomatic process of temporal bone, united by zygomatico temporal suture. This arch leaves a gap from side of skull. Its lateral surface is subcutaneous, medial & lower surfaces give origin to Masseter muscle, while upper border give attachment to temporal fascia.
- 3) External Acoustic Meatus; it lies just below the post. part of post. root of zygoma.



THE SKULL



L/ 1.

Dr. Munqith Mazin

- 4) Supra meatal Triangle (McEwen's triangle); it lies postero-superior to E.A.M., it is bounded by supra mastoid crest above, behind by tympanic plate. This triangle form lateral wall of mastoid antrum.
- 5) Mastoid Part of Temporal Bone; it lies behind E.A.M. It articulates with parietal bone at parietomastoid suture, it also articulates with squamous part of occipital bone at occipitomastoid suture.

***Asterion:** is meeting point of lambdoid, parietomastoid & occipitomastoid sutures. It is called (Postero lateral Fontanel).

***Mastoid Process:** is a large projection from lower part of mastoid bone, postero inferior to E.A.M., It gives attachment to 3 muscles:

1-Sterno mastoid. 2- Splenius Capitis. 3- Longissimus Capitis.

***Styloid Process:** it is a thin, long projection antero-medial to mastoid process, it is directed downward, forward & medially. Its base is ensheathed by tympanic plate and it is related to facial nerve laterally. While its apex is hidden by ramus of mandible.

Styloid process gives attachment to 5 structures:

- i. Stylohyoid muscle
 - ii. Styloglossus muscle
 - iii. Stylopharyngeus muscle
 - iv. Stylohyoid ligament
 - v. Stylomandibular ligament
- 6) **Temporal Fossa:** it is a shallow depression on the side of skull, has following boundaries:
- a) Superiorly: superior temporal line.
 - b) Inferiorly: zygomatic arch laterally, greater wing of sphenoid medially.
 - c) Anterior wall: zygomatic, frontal & sphenoid bones.
Temporal fascia attached to superior temporal line above & to upper border of zygomatic arch below.
 - d) Floor: possess the following:
 1. Temporalis Ms. arise from the whole of temporal fossa.
 2. There is H- shaped sutures where frontal, parietal, sphenoid & temporal bones joined together. This is called Pterion; it lies (1.5 inch) above zygomatic arch. It is of clinical importance because middle meningeal vein & anterior division of middle meningeal artery lie beneath it.
 3. Middle Temporal Vessels which produce markings just above E.A.M.



THE SKULL



L/ 1.

Dr. Munqith Mazin

- e) Contents of temporal fossa:
 - 1. Temporalis muscle
 - 2. Deep temporal nerves
 - 3. Deep temporal vessels.
 - 4. Middle temporal vessels
 - 5. Zygomatico temporal vessels.
 - 6. Zygomatico temporal nerve.
- 7) **Infra temporal Fossa**; it is irregular space lies below middle cranial fossa, behind body of maxilla, lateral to lateral pterygoid plate. Its boundaries are:
 - a) Roof: formed by greater wing of sphenoid & small part of squamous temporal bone. Infra temporal fossa is communicated with temporal fossa through gap deep to zygomatic arch. The roof is pierced by foramen ovale & spinosum.
 - b) Floor is open.
 - c) Posterior wall is open.
 - d) Lateral wall is formed by ramus of mandible.
 - e) Medial wall is formed by lateral pterygoid plate.
 - f) Anterior wall is formed by post. surface of maxilla & medial Surface of zygomatic bone. The anterior & medial walls are separated in the upper part by pterygomaxillary fissure (through which infra temporal fossa communicated with pterygopalatine fossa). Also, this fissure communicates infratemporal fossa with the orbit.

Contents of infra temporal fossa:

- 1. Temporalis muscle
 - 2. Lateral pterygoid muscle
 - 3. Medial pterygoid muscle.
 - 4. Buccinator muscle
 - 5. Maxillary artery & its branches.
 - 6. Maxillary vein.
 - 7. Pterygoid venous plexus
 - 8. Mandibular nerve.
 - 9. Chordae tympani.
 - 10. Maxillary nerve
 - 11. Sphenomandibular ligament.
- 8) **Pterygopalatine Fossa**; it is pyramidal space lies deep to pterygomaxillary fissure & below apex of orbit. Its boundaries are:
- a) Anterior wall: is formed by post. surface of maxilla



THE SKULL



L/ 1.

Dr. Munqith Mazin

- b) Posterior wall: is formed by root of pterygoid process & greater wing of sphenoid.
- c) Medial wall: is formed by perpendicular plate of palatine bone
- d) Lateral wall: is open
- e) Roof: greater wing of sphenoid

Communications of Pterygopalatine Fossa:

1. With orbit through inferior orbital fissure
2. With middle cranial Fossa through foramen rotundum
3. With foramen lacerum through pterygoid canal
4. With pharynx through palatovaginal canal
5. With nose through sphenopalatine foramen
6. With infra temporal Fossa through pterygomaxillary fissure
7. With oral cavity through greater & lesser palatine canals

Contents:

1. Third part of maxillary artery
2. Maxillary nerve & its branches
3. Pterygopalatine ganglion

NORMA BASALIS:

It means viewing of skull from inferior aspect. It is bounded by :

- 1- Anteriorly: there is incisor teeth of maxilla
- 2- Posteriorly: there is superior nuchal line
- 3- Laterally: there is zygomatic arch & mastoid process

There are two transverse lines which divide Norma basalis into three parts:

- A- First line passes behind third molar tooth
- B- Second line passes through anterior margin of foramen magnum

These lines divide Norma basalis into 3 parts:

A- Anterior part: it has the following features:

- 1- Alveolar Arch: which bears sockets of roots of upper teeth
- 2- Hard Palate: it is formed by palatine process of maxilla (ant. $\frac{2}{3}$) and horizontal plate of palatine bone (post. $\frac{1}{3}$), these two parts united by cruciform suture which



consists of inter maxillary, inter palatine & palate maxillary sutures. There is a fossa lies anteriorly in the median plane called Incisive fossa, which contain right & left incisive foramen.

Greater palatine foramen lies just behind lateral part of palatomaxillary suture, while lesser palatine foramen (1-3 inch) lies behind greater palatine foramen, it perforates pyramidal process of palatine bone. The posterior border of hard palate is free and possess a posterior nasal spine in the median plane which gives attachment to Musculus Uvulae.

B- Middle part of Norma basalis:

It extends from posterior border of hard palate to transverse line which passing through anterior margin of foramen magnum. In the median area there is a Vomer which separates the two posterior nasal apertures, its inferior border articulates with hard palate, while its superior border splits into two alae to articulate with sphenoid.

The lateral border of each ala of vomer comes in relation with vaginal process of medial pterygoid plate to form a canal called Vomerovaginal canal. Behind vomer there is bony bar formed by fusion of posterior part of body of sphenoid and basilar part of occipital bone. In front of foramen magnum there is a pharyngeal tubercle which give insertion of superior constrictor, anterior to it forms roof of nasopharynx, lateral to pharyngeal tubercle there is longissimus Capitis inserted, posterior to it Rectus Capitis is inserted. Lateral Area of Middle part of N.B.: Just behind third molar tooth there is pterygoid process (it represents junction of greater wing of sphenoid & body of sphenoid).

This process divided inferiorly into medial & lateral pterygoid plates, which are fused anteriorly but separated posteriorly by pterygoid fossa (which contain deep head of medial pterygoid & tensor palati Ms.)

Medial Pterygoid Plate: it is directed backward, its lateral surface is related to tensor palati Ms., while medial surface form lateral wall of nasopharynx. Its posterior border has the following features:

- 1- Pharyngo-basilar fascia attach to it.
- 2- There is scaphoid fossa which give origin to tensor palati muscle.
- 3- There is pterygoid hamulus which give origin to superior constrictor & pterygomandibular raphe.



Lateral Pterygoid Plate: it is directed backward & laterally, its lateral surface form medial wall of infra temporal fossa, give origin to lower head of lateral pterygoid Ms. While medial surface give origin to deep head of medial pterygoid Ms. Posterolateral to upper end of posterior border of lateral pterygoid plate there is foramen ovale (oval in shape). Posterolateral to foramen ovale there is small circular foramen spinosum (spine of sphenoid posterolateral to it), it gives attachment to Sphenomandibular ligament & tensor palati Ms.

Between greater wing and basilar part of occipital bone there is petrous part of temporal bone which is triangular in shape, its apex directed forward & medially and perforated by carotid canal and separated from sphenoid by foramen lacerum (short, wide canal 1 cm. long).

Tympanic Plate (tympanic part of temporal bone) is a triangular plate lies in the angle between petrous & squamous part, it has the following parts:

- 1- Apex; which is directed medially and close to spine of sphenoid
- 2- Base; which is curved & free
- 3- Anterior surface; forms posterior wall of mandibular fossa
- 4- Posterior surface; forms anterior wall, floor, part of posterior wall of E.A.M.
- 5- Upper border; forms petrotympanic fissure
- 6- Lower border; is sharp & free

Squamous Part of Temporal Bone; it forms anterior part of mandibular fossa which articulate with head of mandible to form temporomandibular joint T. M.J. Also, squamous part forms part of roof of infratemporal fossa.

C- Posterior Part of N. B.:

The most important area in this region is foramen magnum which is oval, large foramen wide posteriorly than anteriorly. On each side of anterior part of foramen magnum there is occipital condyle which is oval in shape, it articulates with superior articular facets of atlas to form atlantooccipital joint.

Anterosuperior to occipital condyle there is Hypoglossal canal (Anterior condylar canal). While posterior condylar canal lies behind occipital condyle.

At the posterior margin of foramen magnum started External Occipital Crest continue backward to external occipital protuberance, it gives attachment to Ligamentum nuchae.



THE SKULL



L/ 1.

Dr. Munqith Mazin

From middle of crest arise inferior nuchal line, above superior nuchal line. The part of occipital bone which lie behind foramen magnum called Squamous part of occipital bone. The area between superior & inferior nuchal lines gives insertion to Semispinalis Capitis medially, Superior Oblique Ms. Laterally. While area below inferior nuchal line give insertion to Rectus Capitis posterior minor medially, Rectus Capitis posterior major laterally.

Jugular Foramen located at posterior end of occipitopetrous suture. Posterior to styloid process there is Stylomastoid Foramen, posterior to it there is mastoid process. Medial to mastoid process there is a notch called Mastoid Notch which give origin to Posterior belly of Digastric Ms. Medial to notch there is occipital groove (for occipital artery)

INTERIOR OF THE SKULL (Cranial Cavity)

A- Internal Surface of Cranial Vault.

B- Internal Surface of Base of Skull:

1- Anterior Cranial Fossa

2- Middle Cranial Fossa

3- Posterior Cranial Fossa

Anterior Cranial Fossa:

It is bounded anteriorly and on both sides by frontal bone, while posteriorly there is lesser wing of sphenoid, anterior clinoid process and anterior margin of sulcus chiasmaticus.

The floor of A.C.F. is formed anteriorly by cribriform plate of ethmoid, this plate articulates anteriorly with frontal bone, leaving a blind foramen called foramen caecum, while posteriorly it articulates with superior surface of anterior part of body of sphenoid (Jugum Sphenoidale) leaving posterior ethmoidal canal.

*Cribriform plate separate A.C.F. from nasal cavity

* Jugum Sphenoidale separate A.C.F. from sphenoidal air sinuses. Laterally cribriform plate articulates with orbital plate of frontal bone leaving anterior ethmoidal canal. The cribriform plate shows Crista Galli which projects upward like tooth which provide attachment to Falx Cerebri, the plate is perforated (15 __20 on each side) for passage of olfactory nerve. Lateral to cribriform plate there is orbital



plate of frontal bone which separates A.C.F. from orbit. It lodges frontal lobe of brain, posteriorly it articulates with lesser wing of sphenoid, which has free posterior border to end medially into prominent projection Anterior Clinoid Process which give attachment to free margin of tentorium cerebelli.

Middle Cranial Fossa:

It is deeper than A.C.F., it is bounded by lesser wing of sphenoid, anterior clinoid process & sulcus chiasmaticus anteriorly, while petrous bone & dorsum sellae of sphenoid posteriorly. Laterally there is greater wing of sphenoid, parietal bone & squamous part of temporal bone. Floor of M.C.F. is formed by body of sphenoid in median region which is characterized by:

- 1- Sulcus chiasmaticus which lead on each side to optic canal.
- 2- Optic canal which lead to orbit, boundaries:
 - a- Laterally by lesser wing of sphenoid.
 - b- Medially by body of sphenoid.
 - c- Anteriorly & posteriorly by two roots of lesser wing of sphenoid.
- 3- Sella Turcica: the upper surface of body of sphenoid is hollowed out in front of Turkish saddle called sella turcica which consist of:
 - a- Tuberculum sellae in front, it provides attachment to diaphragma sella, its lateral end form middle clinoid process.
 - b- Hypophyseal fossa in the middle which lodge pituitary gland.
 - c- Dorsum sella behind, which form back of saddle, its superolateral angles expand to form posterior clinoid process (give attachment to tentorium cerebelli).
- 4- Carotid Groove: which is a shallow depression on each side of body of sphenoid extend from foramen lacerum till anterior clinoid process. Superolateral to body of sphenoid there is superior orbital fissure which is bounded by:
 - a- Lesser wing of sphenoid, superiorly.
 - b- Greater wing of sphenoid, inferiorly.
 - c- Body of sphenoid, medially.



THE SKULL



L/ 1.

Dr. Munqith Mazin

Posterior to medial end of superior orbital fissure there is foramen rotundum, posterolateral to foramen rotundum there is foramen ovale, posterolateral to foramen ovale there is foramen spinosum, posteromedial to foramen ovale there is foramen lacerum. Behind foramen lacerum there is trigeminal impression which lodge trigeminal ganglion, lateral to it there is groove for greater petrosal nerve, lateral to it there is groove for lesser petrosal nerve, lateral to it there is arcuate eminence which produced by superior semicircular canal. The superior border of petrous temporale is grooved by superior petrosal sinus, which give attachment of tentorium cerebelli.

Posterior Cranial Fossa:

It is the largest and deepest of the two above. It is bounded by superior border of petrous temporale and dorsum sellae anteriorly, squamous occiput posteriorly, while laterally there is mastoid temporale and parietal bone. This fossa lodges Cerebellum, Pons & Medulla.

Floor: In the median area there is clivus which is a sloping surface in front of foramen magnum (clivus formed by fusion of posterior part of body of sphenoid and basilar part of occipital bone). It supports pons and medulla. This separated from petrous by Petro occipital fissure which is grooved by inferior petrosal sinus. Behind clivus there is foramen magnum which is bounded by Basi occiput anteriorly, squamous temporale posteriorly, condylar part of occipital bone laterally. Behind foramen magnum there is squamous part of occipital bone, in the median plane there is prominent bony ridge called internal occipital protuberance which match external occipital protuberance.